

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A

FD

BC

0607

DE

0004

HL

0014

PSW

0000

PC

420C

SP

FFFF

Int-Reg

00

Flag

S

1

Z

0

AC

0

P

0

C

1

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Load me at

1 LDA 8000

2 MOV B,A

3 LDA 8001

4 SUB B

5 STA 8002

6 RST 1

DataStackKeyPadMemoryI/O Ports

Start8000OK

Address (Hex)	Address	Data
1F40	8000	6
1F41	8001	3
1F42	8002	3
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

FileResetAssemblerDebugHelp

Registers

A0B

BC0900

DE0000

HL0000

PSW0000

PC420C

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LDA 8050

2MOV B,A

3LDA 8051

4ADD B

5STA 8052

6HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

1F7280509

1F7380512

1F74805211

1F7580530

1F7680540

1F7780550

1F7880560

1F7980570

1F7A80580

1F7B80590

1F7C80600

1F7D80610

1F7E80620

1F7F80630

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File

Reset

Assembler

Debug

Help

Registers

Flag

Load me at

A

05

S

0

BC

05

00

Z

1

DE

00

04

AC

0

HL

00

14

P

1

PSW

00

00

C

0

PC

42

16

SP

FF

FF

Int-Reg

00

1

LDA

2200

2

MOV

E,A

3

MVI

D,00

4

LDA

2201

5

MOV

C,A

6

LXI

H,0000

7

BACK:

DAD

D

8

DCR

C

9

JNZ

BACK

10

SHLD

2202

11

HLT

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Data

Stack

KeyPad

Memory

I/O Ports

Start

2200

OK

Address (Hex)

Address

Data

0898

2200

4

0899

2201

5

089A

2202

20

089B

2203

0

089C

2204

0

089D

2205

0

089E

2206

0

089F

2207

0

08A0

2208

0

08A1

2209

0

08A2

2210

0

08A3

2211

0

08A4

2212

0

08A5

2213

0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

FileResetAssemblerDebugHelp

Registers

A07

BC0307

DE0004

HL0014

PSW0000

PC421B

SPFFFF

Int-Reg00

Flag

S1

Z0

AC0

P0

C1

Decimal - Hex Conversion

Decimal

Hex

00

00

To Hex

To Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

GNUSim8085 - 8085 Microprocessor Simulator

Load me at

1START: NOP

2LDA 8500

3MOV B,A

4LDA 8501

5MVI C,00H

6

7LOOP: CMP B

8JC LOOP1

9SUB B

10INR C

11JMP LOOP

12

13LOOP1: STA 8502

14MOV A,C

15STA 8503

16HLT

DataStackKeyPadMemoryI/O Ports

Start8500OK

Address (Hex)	Address	Data
2134	8500	3
2135	8501	21
2136	8502	0
2137	8503	7
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0
2140	8512	0
2141	8513	0

Line No	Assembler Message
0	Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A0B

BC0607

DE0004

HL0014

PSW0000

PC4217

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

DecimalHex

00

0

To Hex

To Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LDA3050

2MOV B,A

3LDA3051

4ADD B

5STA3052

6LDA3053

7MOV B,A

8LDA3054

9ADC B

10STA3055

11HLT

12

DataStackKeyPadMemoryI/O Ports

Start3050OK

Address (Hex)AddressData

0BEA30502

0BEB30513

0BEC30525

0BED30536

0BEE30545

0BEF305511

0BF030560

0BF130570

0BF230580

0BF330590

0BF430600

0BF530610

0BF630620

0BF730630

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

FileResetAssemblerDebugHelp

Registers

A00

BC0600

DE0002

HL0003

PSW0000

PC4214

SPFFFF

Int-Reg00

S0

Z1

AC0

P1

C0

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Load me at

1LHLD2050

2XCHG

3LHLD2052

4MVI C,00

5MOV A,E

6SUB L

7STA2054

8MOV A,D

9SUB H

10STA2055

11HLT

12

13

DataStackKeyPadMemoryI/O Ports

Start2050OK

Address (Hex)	Address	Data
0802	2050	2
0803	2051	0
0804	2052	3
0805	2053	0
0806	2054	1
0807	2055	1
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A00

BC0000

DE0000

HL0000

PSW0000

PC4222

SP000A

Int-Reg00

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Load me at

1LHLD 2050

2SPHL

3LHLD 2052

4XCHG

5LXI H,0000H

6LXI B,0000H

7AGAIN: DAD SP

8JNC START

9INX B

10START: DCX D

11MOV A,E

12ORA D

13JNZ AGAIN

14SHLD 2054

15MOV L,C

16MOV H,B

17SHLD 2055

18HLT

19

20

21

DataStackKeyPadMemoryI/O Ports

Start2050OK

Address (Hex)AddressData

0802205010

080320510

080420525

080520530

0806205450

080720555

080820560

080920570

080A20580

080B20590

080C20600

080D20610

080E20620

080F20630

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A02

BC1E00

DE0000

HL0000

PSW0000

PC421B

SPFFFF

Int-Reg00

Flag

S1

Z0

AC0

P1

C1

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LDA 8501

2MOV B,A

3LDA 8500

4MVI C,00

5

6LOOP: CMP B

7JC LOOP1

8SUB B

9INR C

10JMP LOOP

11

12STA 8503

13DCR C

14MOV A,C

15LOOP1: STA 8502

16RST 1

DataStackKeyPadMemoryI/O Ports

Start8500OK

Address (Hex)AddressData

213485002

2135850130

2136850215

213785032

213885040

213985050

213A85060

213B85070

213C85080

213D85090

213E85100

213F85110

214085120

214185130

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle